

Q1. What is the expansion of $(x + 2)^2$?

- (A) $x^2 + 4x + 4$
 (B) $x^2 + 2x + 4$
 (C) $x^2 - 4x + 4$
 (D) $x^2 + 4$
 (E) $x^2 - 4$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 2x + 1$
 (B) $x^2 - 3x - 2$
 (C) $x^2 + 5x + 6$
 (D) $x^2 - 2x - 3$
 (E) $x^2 + x + 1$

Q3. What is the factored form of $x^2 + 6x + 9$?

- (A) $(x + 3)^2$
 (B) $(x - 3)^2$
 (C) $(x + 3)(x - 3)$
 (D) $(x + 2)(x + 4)$
 (E) $(x - 2)(x - 4)$

Q4. Which of the following is the expansion of $(x - 3)^2$?

- (A) $x^2 - 6x + 9$
 (B) $x^2 + 6x + 9$
 (C) $x^2 - 9$
 (D) $x^2 + 9$
 (E) $x^2 - 6x - 9$

Q5. What is the factored form of $x^2 - 8x + 16$?

- (A) $(x + 4)^2$
 (B) $(x - 4)^2$
 (C) $(x + 2)(x - 6)$
 (D) $(x - 2)(x + 6)$
 (E) $(x + 8)(x - 2)$

Q6. Which of the following is a perfect square trinomial?

- (A) $x^2 + 2x - 3$
 (B) $x^2 - 4x + 4$
 (C) $x^2 + x - 2$
 (D) $x^2 - 2x - 3$
 (E) $x^2 + 3x + 2$

Q7. What is the expansion of $(x + 5)^2$?

- (A) $x^2 + 10x + 25$
 (B) $x^2 - 10x + 25$
 (C) $x^2 + 5x + 25$
 (D) $x^2 - 5x + 25$
 (E) $x^2 + 25$

Q8. Which of the following is the factored form of $x^2 + 12x + 36$?

- (A) $(x + 6)^2$
 (B) $(x - 6)^2$
 (C) $(x + 3)(x + 9)$
 (D) $(x - 3)(x - 9)$
 (E) $(x + 2)(x + 10)$

Open Ended Questions (FRQ)**Q9.** Factor the expression $x^2 + 10x + 25$. Show your work and explain your reasoning.**Q10.** Expand the expression $(x - 4)^2$ and explain why it is a perfect square trinomial.**Chef's Tips**

- Tip 1: Ensure students understand the concept of perfect square trinomials.
- Tip 2: Emphasize the importance of factoring and expanding expressions correctly.
- Tip 3: Encourage students to show their work and explain their reasoning for free response questions.